

Notes: Lexical (n-gram) Metrics for Hallucination Detection

1. Background and Motivation

In Natural Language Generation (NLG), evaluating whether generated text is **faithful** to a source or reference is critical. Early and widely used evaluation metrics such as **ROUGE**, **BLEU**, and **METEOR** rely on **lexical overlap**, i.e., similarity at the word or phrase level. These metrics are simple, fast, and correlate reasonably with human judgments of fluency, but they have serious limitations when used to detect **hallucinations**.

Hallucination refers to content that is fluent but **unsupported, incorrect, or contradictory** with respect to the source or reference.

2. Lexical Features and n-grams

Lexical features

Lexical features operate at the surface level of text and include: - Words - Phrases - Fixed word sequences

They do not capture deep semantics or world knowledge.

n-grams

An **n-gram** is a contiguous sequence of n tokens.

Example sentence:

"Alan Turing was a British mathematician"

- Unigrams ($n = 1$):
 - Alan, Turing, was, a, British, mathematician
- Bigrams ($n = 2$):
 - Alan Turing, Turing was, was a, a British, British mathematician
- Trigrams ($n = 3$):
 - Alan Turing was, Turing was a, was a British, a British mathematician

Metrics like ROUGE and BLEU count how many n-grams in the generated text also appear in the reference text.

3. Measuring Information Overlap with n-grams

Core idea

If a generated text shares many n-grams with the reference, it is assumed to convey similar information.

Example: High overlap (faithful generation)

Reference text:

"Alan Turing was a British mathematician."

Generated text:

"Alan Turing was a famous British mathematician."

Overlapping bigrams: - Alan Turing - Turing was - was a - British mathematician

Result: - High n-gram overlap - High ROUGE / BLEU score - Considered faithful by lexical metrics

4. Using n-grams to Approximate Hallucination

Key assumption

n-grams that appear in the generated text but do not appear in the source or reference may indicate hallucination.

This is only an approximation: the metrics cannot truly reason about facts, but they penalize **unsupported lexical content**.

5. Same Example: Faithful vs Hallucinated Output

Reference text

"Alan Turing was a British mathematician."

Case A: Faithful but extended generation

Generated text:

"Alan Turing was a British mathematician and logician."

Observations: - "logician" does not appear in the reference - However, it does not contradict the reference - n-gram overlap remains high

Metric behavior: - Slight penalty for extra content - Still mostly rewarded

Case B: Hallucinated generation

Generated text:

"Alan Turing was a German physicist."

New bigrams introduced: - German physicist - a German - was a German

Observations: - These n-grams do not appear in the reference - They introduce incorrect factual information

Metric behavior: - Sharp drop in n-gram overlap - Hallucination indirectly penalized

6. Same Example Showing a Critical Failure of n-grams

Reference text:

"Paris is the capital of France."

Generated text:

"Paris is the capital of Germany."

Overlapping n-grams: - Paris is - is the capital of

Only one word differs: - France → Germany

Observations: - Lexical overlap is still very high - Meaning is completely wrong

Conclusion: - n-gram metrics fail to detect semantic contradiction - This is a classic hallucination that passes ROUGE/BLEU

7. Why Lexical Metrics Are Insufficient for Hallucination Detection

- They measure **surface similarity**, not truth
- They cannot reason about:
 - Entity-attribute binding
 - World knowledge

- Logical contradiction
 - High scores can coexist with severe hallucinations
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8. Why Metrics Like PARENT Were Proposed

PARENT extends lexical matching by: - Comparing generated text with both: - Source data (e.g., tables / infoboxes) - Reference text - Computing precision and recall of supported n-grams

However: - Still relies on lexical overlap - Cannot fully solve hallucination

9. Key Takeaways

- n-gram overlap approximates information similarity, not factual correctness
 - Unsupported or contradictory facts may evade lexical metrics
 - Dataset evaluation based purely on ROUGE/BLEU encourages hallucination
 - This limitation motivates semantic, entailment-based, and internal-state-based hallucination detection methods
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10. One-line Summary

Lexical n-gram metrics reward similarity in wording, not faithfulness in meaning, making them unreliable for detecting hallucinations.